

Toward workforce pathways into the semiconductor manufacturing sector

Specific Aims

Aim 1: Advance workforce pathways into the semiconductor manufacturing sector via a population-focused strategy.

Aim 2: Advance workforce pathways into the semiconductor manufacturing sector via a population-focused strategy.

Background

Our preliminary data suggest a tractable path toward measurable improvement. Findings will be disseminated through peer-reviewed publication and open data release. Rigorous, pre-registered analyses will guard against bias and support reproducibility. This proposal addresses workforce pathways into the semiconductor manufacturing sector, a problem with substantial significance.

Approach

This proposal addresses workforce pathways into the semiconductor manufacturing sector, a problem with substantial significance. Prior work has established a partial understanding, yet key gaps remain. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Findings will be disseminated through peer-reviewed publication and open data release.

Innovation

Rigorous, pre-registered analyses will guard against bias and support reproducibility.

Investigator Context

Two co-equal PIs contribute complementary methods under joint leadership. The R2 host institution offers focused departmental resources and regional research partnerships, backed by a growing institutional research base.

Budget Justification

Total requested support is \$514,158 over 3 years at a R2 institution, covering personnel, materials, and dissemination.

References

1. [1] <https://doi.org/10.1145/3292500.3330701>
2. [2] <https://doi.org/10.1371/journal.pone.0173664>
3. [3] <https://doi.org/10.1371/journal.pone.0173664>

4. [4] <https://doi.org/10.1093/nar/gky1055>
5. [5] <https://doi.org/10.1002/anie.201907688>
6. [6] <https://doi.org/10.1126/science.1259855>
7. [7] <https://doi.org/10.1056/NEJMoa2034577>